

Problems in Plane Geometry

A Collection of Problems

*From Side–Angle Inequalities to Common
Theorems — Similarity, Pythagoras, BPT, Circles,
Quadrilaterals, Ceva, Menelaus, Ptolemy and Beyond*

Throughout this collection, a, b, c denote the side-lengths of $\triangle ABC$ opposite to $\angle A, \angle B, \angle C$ respectively, s denotes the semiperimeter, R the circumradius, r the inradius, and $[\triangle ABC]$ the area of $\triangle ABC$, unless mentioned otherwise. All angles are measured in degrees unless specified.

The collection is organised into eight parts. Parts 1–2 develop the foundational material on inequalities, congruence and elementary configurations. Parts 3–8 develop the major themes of competitive exam level Euclidean geometry — similarity and the Basic Proportionality Theorem, the Pythagorean theorem and length identities, quadrilaterals, circle geometry and Ptolemy, the theorems of Ceva and Menelaus, and finally a tour of triangle centres and advanced topics.

Part 1 — Triangle and Side–Angle Inequalities

1. Suppose P is any point inside $\triangle ABC$ and s is the semiperimeter of $\triangle ABC$, that is, $AB + BC + CA = 2s$. Prove that $s < AP + BP + CP < 2s$.
2. Let D be any point on the side BC of an equilateral $\triangle ABC$. E is the midpoint of AD . Show that $AE < CE$.
3. In $\triangle ABC$, straight lines are drawn from the vertices B, C to intersect at a point O within the triangle. Prove that
(a) $AB + AC > OB + OC$, (b) $\angle BOC > \angle BAC$, (c) $AB + AC < OB + OC + 2 \cdot OA$.
4. Let a, b, c be the lengths of sides of $\triangle ABC$, and $\angle A, \angle B, \angle C$ be the measures of the corresponding angles. Prove that

$$60^\circ \leq \frac{a \cdot \angle A + b \cdot \angle B + c \cdot \angle C}{a + b + c} < 90^\circ.$$

5. In $\triangle ABC$, M is the midpoint of BC . If $AM > \frac{BC}{2}$, then prove that $\angle BAC$ is acute.
6. Suppose that M is the midpoint of BC in $\triangle ABC$. Prove that $AM > \frac{1}{2}(AB + AC - BC)$.
7. In $\triangle ABC$, the bisector of $\angle BAC$ meets BC at D . Prove that $AB > BD$ and $AC > CD$.
8. In $\triangle ABC$, $AB > AC$. Let D be any point on BC . Prove that $AB > AD$.
9. In $\triangle ABC$, $AD \perp BC$, $BE \perp AC$ and $CF \perp AB$. Prove that $AD + BE + CF < AB + BC + CA$.
10. $ABCD$ is a convex quadrilateral and $AB + BD < AC + CD$. Prove that $AB < AC$.

11. In $\triangle ABC$, $AB > AC$. E, F are the midpoints of AC and AB respectively. Show that $BE > CF$.

12. In $\triangle ABC$, $AB > AC$. M is the midpoint of BC . Prove that $\angle MAB < \angle MAC$.

13. Let P be a fixed point outside a circle centred at O . Find a point M on the circumference of the circle so that the distance PM is minimum. (What about the position of point M so that PM is maximum?)

14. In $\triangle ABC$, $AB > AC$. The bisector of $\angle BAC$ meets BC at D . Prove that $BD > CD$.

15. Prove that the distance between any two points inside a triangle cannot be greater than its semiperimeter.

16. Square $ABCD$ is divided into two parts by the diagonal AC . Prove that, if O is any point within $\triangle ABC$, then $OB < OD$.

17. Let D, E, F be the midpoints of sides BC, CA, AB respectively in $\triangle ABC$. Prove that

$$AD + BE + CF < AB + BC + CA < \frac{4}{3}(AD + BE + CF).$$

18. In $\triangle ABC$, $AB > AC$. The bisectors of $\angle ABC$ and $\angle ACB$ meet AC and AB at E and F respectively. Prove that $BE > CF$.

19. Using **18** or otherwise, prove that if the two angle bisectors of a triangle have equal lengths, then the triangle is isosceles. (*Steiner–Lehmus Theorem*)

20. If the lengths of a triangle's sides satisfy the inequality $a^2 + b^2 > 5c^2$, then prove that c is the length of the shortest side.

21. Let a, b, c be sides of a triangle. Prove that $a^2 + b^2 + c^2 < 2(ab + bc + ca)$.

From now on, a, b, c will denote the lengths of the sides of a triangle, unless mentioned otherwise.

22. Prove that

$$a(b - c)^2 + b(c - a)^2 + c(a - b)^2 + 4abc > a^3 + b^3 + c^3.$$

23. Let $p = \frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ and $q = \frac{a}{c} + \frac{c}{b} + \frac{b}{a}$. Prove that $|p - q| < 1$.

24. Prove that $(a + b - c)(a - b + c)(b + c - a) \leq abc$.

25. Prove that $a^2b(a - b) + b^2c(b - c) + c^2a(c - a) \geq 0$.

26. Prove that $a^2 + b^2 + c^2 > \sqrt{3} \max\{|a^2 - b^2|, |b^2 - c^2|, |c^2 - a^2|\}$.

27. Prove that $3(ab + bc + ca) \leq (a + b + c)^2 \leq 4(ab + bc + ca)$.

28. Let s be the semiperimeter of a triangle, that is, $a + b + c = 2s$. Prove that

$$a^2 + b^2 + c^2 \geq \frac{36}{35} \left(s^2 + \frac{abc}{s} \right).$$

From now on, s will denote the semiperimeter of a triangle with side-lengths a, b, c .

29. Prove that $8(s-a)(s-b)(s-c) \leq abc$.

30. Prove that

$$\frac{3}{2} \leq \frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} < 2.$$

31. Using **30** or otherwise prove that

$$\frac{15}{4} \leq \frac{s+a}{b+c} + \frac{s+b}{c+a} + \frac{s+c}{a+b} < \frac{9}{2}.$$

32. Prove that $\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} \geq \frac{9}{s}$.

33. Prove that $\frac{2s}{abc} \leq \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$.

34. Prove that

$$\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \geq 3.$$

35. Let x, y, z be three distinct real positive numbers. Determine with proof whether or not the three real numbers

$$\left| \frac{x}{y} - \frac{y}{x} \right|, \quad \left| \frac{y}{z} - \frac{z}{y} \right|, \quad \left| \frac{z}{x} - \frac{x}{z} \right|$$

can be the lengths of the sides of a triangle.

36. Prove that $\frac{1}{3} \leq \frac{a^2 + b^2 + c^2}{(a+b+c)^2} < \frac{1}{2}$.

37. Prove that

$$abc < a^2(s-a) + b^2(s-b) + c^2(s-c) \leq \frac{3}{2} abc.$$

38. Prove that

$$\left| \frac{a-b}{a+b} + \frac{b-c}{b+c} + \frac{c-a}{c+a} \right| < \frac{1}{8}.$$

Part 2 — Foundational Problems and Congruence

39. In $\triangle ABC$, $\angle C = 90^\circ$. A square $ABPQ$ is drawn on the side AB . The perpendicular from P to AC meets AC at N . Prove that $PN = AC + BC$.

40. In a square $ABCD$, M and N are the midpoints of the sides AB and BC respectively. Prove that $AN \perp DM$.

41. Prove that, if in any triangle the median on a certain side is half the length of that particular side, then the triangle is right angled.

42. In any $\triangle ABC$, let E and D be points on AC and BC , respectively. The bisectors of $\angle CAD$ and $\angle CBE$ meet each other at F . Prove that $\angle AEB + \angle ADB = 2\angle AFB$.

43. In $\triangle ABC$, let D be a point on AC such that $AB = AD$. If $\angle ABC - \angle ACB = 30^\circ$, find $\angle CBD$.

44. In $\triangle ABC$, $BC = 2AB$. D is the midpoint of BC , and E is the midpoint of BD . Prove that AD bisects $\angle CAE$.
45. Let X be the midpoint of side AB of $\triangle ABC$. Let Y be the midpoint of CX . Let BY cut AC at Z . Prove that $AZ = 2 \cdot ZC$.
46. Let D be any point on the side AC of $\triangle ABC$. P, Q, X and Y are midpoints of AB, BC, AD and DC respectively. Prove that $PX = QY$.
47. In trapezium $ABCD$, $AB \parallel CD$. Let P and Q be the midpoints of AD and BC respectively. Prove that $PQ \parallel CD$ and $PQ = \frac{1}{2}(AB + CD)$.
48. In $\triangle ABC$, the medians CD and BE are extended to P and T , respectively, so that $CD = DP$ and $BE = ET$. Prove that the points P, A and T are collinear and A is the midpoint of PT .
49. In $\triangle ABC$, let BM be the external bisector of $\angle B$, and CN be the external bisector of $\angle C$. Draw $AE \perp BM$ and $AF \perp CN$. Prove that $EF \parallel BC$ and $EF = \frac{1}{2}(AB + BC + CA)$.
50. Prove that the quadrilateral obtained by joining the midpoints of any given (convex) quadrilateral is always a parallelogram.
51. In any $\triangle ABC$, D, E and F are the midpoints of sides AC, AB and BC , respectively. Draw $BG \perp AC$. Prove that $\angle EGF = \angle EDF$.
52. $\triangle ABC$ is isosceles with $AB = AC$. The base BC is extended on either side to D and E so that $BD = CE$. Prove that $AD = AE$.
53. $\triangle ABC$ is isosceles with $\angle B = \angle C = 2\angle A$. If the bisector of $\angle B$ meets AC at D , prove that $AD = BC$.
54. In $\triangle ABC$, $AB = AC$. If D is on extended BC such that $BC = CD$ and E is on extended AB , so that $AB = BE$, prove that $AD = CE$.
55. Let D and E be points on sides AC and AB respectively of $\triangle ABC$. If $AD = AE$ and $BD = CE$, prove that $AB = AC$.
56. The internal bisector of $\angle B$, and the external bisector of $\angle C$ of $\triangle ABC$ meet at D . Through D , a line parallel to CB meets AC at L and AB at M . Prove that $LM = MB - LC$.
57. Let E be a point inside a square $ABCD$ such that $\angle EDC = \angle ECD = 15^\circ$. Prove that $\triangle ABE$ is equilateral.
58. Given $\triangle ABC$, draw squares $ABDE$ and $CAFG$ outwards with bases AB, CA respectively. Let M be the midpoint of BC . Prove that $AM = \frac{1}{2}EF$.
59. In $\triangle ABC$, D is the midpoint of BC . E is a point on AC such that BE intersects AD at P and $BP = AC$. Prove that $AE = PE$.
60. In $\triangle ABC$, median AM is such that $\angle MAC = 2\angle MAB$. AM is extended through M to D so that $\angle DBA$ is a right angle. Prove that $AC = \frac{1}{2}AD$.
61. Given $\triangle ABC$ where $\angle B = 2\angle C$. D is a point on BC such that AD bisects $\angle A$. Prove that $AC = AB + BD$.

- 62.** Let $ABCD$ be a square. E, F are points on BC, CD respectively and $\angle EAF = 45^\circ$. Prove that $EF = BE + DF$.
- 63.** Given any $\triangle ABC$, AE bisects $\angle BAC$, BD bisects $\angle ABC$. Draw $CP \perp BD$, and $CQ \perp AE$. Prove that PQ is parallel to AB .
- 64.** In $\triangle ABC$, $\angle C = 60^\circ$. The bisector of $\angle A$ meets BC at P , and the bisector of $\angle B$ meets AC at Q . Show that $AB = AQ + BP$.
- 65.** $\triangle ABC$ is isosceles with $AB = AC$, and $\angle A = 120^\circ$. If the base BC is trisected at D and E (that is, D and E are points on BC such that $BD = DE = EC$), then prove that $\triangle ADE$ is equilateral.
- 66.** On BC as a base, an equilateral $\triangle ABC$ and an isosceles $\triangle DBC$ are drawn on the same side of BC such that $\angle D = \frac{1}{2}\angle A$. Prove that $AD = BC$.
- 67.** The side AB of an equilateral $\triangle ABC$ is extended to D such that $BD = 2 \cdot AB$. A perpendicular DF is drawn from D to extended CB . Show that $\angle FAC = 90^\circ$.
- 68.** In $\triangle ABC$, $\angle C = 90^\circ$. D and E are points on AB so that $BD = BC$ and $AE = AC$. Draw $EF \perp BC$ and $DG \perp AC$. Prove that $DE = EF + DG$.
- 69.** In square $ABCD$, AC and BD intersect at E . CF bisects $\angle ACD$. The perpendicular from B to CF cuts AC at P and CD at Q . Prove that $DQ = 2 \cdot PE$.
- 70.** Let BC be a fixed line segment. PQ is a line parallel to BC . Prove that, among all triangles with the same base BC , and vertices lying on the line PQ , the one with the least perimeter is isosceles.

Part 3 — Similarity and the Basic Proportionality Theorem

- 71.** In $\triangle ABC$, D is on BC such that $\angle ADB = \angle BAC$. Prove that $AB^2 = BD \cdot BC$, and deduce that AB is tangent at A to the circumcircle of $\triangle ADC$.
- 72.** In $\triangle ABC$, the bisector of $\angle A$ meets BC at D and the circumcircle again at M . Prove that $MA \cdot MD = MB^2 = MC^2$, and that $AD \cdot AM = AB \cdot AC$.
- 73.** In $\triangle ABC$, D, E, F are points on BC, CA, AB such that AD, BE, CF are concurrent at P . A line through P parallel to BC meets CA, AB at X, Y . Prove that $PX = PY$.
- 74.** In a trapezium $ABCD$ with $AB \parallel CD$, the diagonals meet at O . A line through O parallel to AB meets the legs AD, BC at P, Q . Prove that $\frac{2}{PQ} = \frac{1}{AB} + \frac{1}{CD}$, and that PQ is the harmonic mean of AB and CD .
- 75.** In $\triangle ABC$, points D, E lie on BC , and F, G lie on AC , with $DF \parallel AB$ and $EG \parallel AB$. The lines DG and EF meet at P . Prove that AP bisects DE .
- 76.** $ABCD$ is a parallelogram. A line through A meets BD at P , BC at Q , and the extension of DC at R . Prove that $AP^2 = PQ \cdot PR$.
- 77.** In $\triangle ABC$, D is the midpoint of BC . A line through D meets AB at E and AC extended at F . Prove that $\frac{AE}{EB} = \frac{AF}{FC}$.

- 78.** Let $ABCD$ be a square. E lies on BC and F on CD with $\angle EAF = 45^\circ$. AE and AF meet BD at P and Q respectively. Prove that $\triangle APQ \sim \triangle AFE$ and that $EF^2 = 2 \cdot PQ \cdot EF$.
- 79.** In $\triangle ABC$, D is on AB , E on AC with $DE \parallel BC$. The diagonals of trapezium $BDEC$ meet at P . Prove that A, P and the midpoint of BC are collinear.
- 80.** In $\triangle ABC$, the cevian AD from A to BC is divided by a line ℓ in the ratio λ (counting from A). The line ℓ meets AB and AC at E and F respectively. Prove that $\frac{AE}{EB} + \frac{AF}{FC}$ depends only on λ , and find its value.
- 81.** In $\triangle ABC$, the angle bisector from A meets BC at D . Prove $AD^2 = AB \cdot AC - BD \cdot DC$.
- 82.** In $\triangle ABC$, points D on BC and E on CA are such that $AD = AE$ and $\angle BAD = \angle CAE$. Prove that $\triangle ABD \sim \triangle ACE$ if and only if $\triangle ABC$ is isosceles with $AB = AC$, and otherwise determine the relation between $AB \cdot AE$ and $AC \cdot AD$.
- 83.** In $\triangle ABC$, the median AM to BC meets the circumcircle at N . Prove that $AM \cdot MN = BM \cdot MC = \frac{a^2}{4}$, and deduce $AM \cdot AN = AB \cdot AC$.
- 84.** A line ℓ parallel to side BC of $\triangle ABC$ cuts AB, AC at D, E and divides the triangle into two regions of equal area. Prove that $\frac{AD}{AB} = \frac{1}{\sqrt{2}}$ and $\frac{[\triangle ADE]}{[BDEC]} = 1$. Generalise: a line parallel to BC divides the area in ratio $\lambda : 1 - \lambda$ iff $\frac{AD}{AB} = \sqrt{\lambda}$.
- 85.** In $\triangle ABC$, three concurrent cevians AD, BE, CF meet at P . A line through P meets sides AB, AC at X, Y . Prove

$$\frac{AX}{XB} \cdot \frac{BD}{DC} + \frac{AY}{YC} \cdot \frac{CD}{DB} = \frac{AP}{PD}.$$

Part 4 — Pythagoras Theorem and Length Identities

- 86.** (*Apollonius' Theorem*) In $\triangle ABC$, let AD be the median to BC . Prove $AB^2 + AC^2 = 2(AD^2 + BD^2)$, and deduce that the sum of the squares of the medians of any triangle equals $\frac{3}{4}(a^2 + b^2 + c^2)$.
- 87.** (*Stewart's Theorem*) In $\triangle ABC$, D lies on BC with $BD = m$, $DC = n$, $AD = d$. Prove

$$AB^2 \cdot n + AC^2 \cdot m - AD^2 \cdot a = amn, \quad a = m + n.$$

- 88.** Use Stewart's Theorem to prove that in $\triangle ABC$ the length of the internal bisector from A is

$$t_a^2 = bc \left[1 - \left(\frac{a}{b+c} \right)^2 \right],$$

and derive the analogous formula for the length of the median.

- 89.** (*British Flag Theorem*) In a rectangle $ABCD$, P is any point in the plane. Prove $PA^2 + PC^2 = PB^2 + PD^2$, and prove the converse: if P is a point such that this identity holds for the four vertices of a quadrilateral $ABCD$, then $ABCD$ has perpendicular diagonals and lines AC, BD pass through a common foot from P .

90. $ABCD$ is a square and P is a point inside it such that $PA = 1$, $PB = 2$, $PC = 3$. Find $\angle APB$ and the side of the square.

91. In $\triangle ABC$, $\angle A = 90^\circ$ and AD is the altitude to the hypotenuse. Prove that $\frac{1}{AD^2} = \frac{1}{AB^2} + \frac{1}{AC^2}$, and that the inradius r and the altitude h_a satisfy $r = \frac{AB + AC - BC}{2}$, $h_a = \frac{AB \cdot AC}{BC}$.

92. A right $\triangle ABC$ has legs b, c and hypotenuse a . Squares are erected externally on each side. Prove that the centroid of the union of the three squares lies on the perpendicular from the right-angle vertex to the hypotenuse, and find its distance from the right-angle vertex in terms of a, b, c .

93. (*Generalised Pythagoras / Law of Cosines*) Prove that in any $\triangle ABC$, $c^2 = a^2 + b^2 - 2ab \cos C$, and use it to give a one-line proof that the locus of points P with $PB^2 - PC^2$ constant is a line perpendicular to BC .

94. P is a point inside an equilateral $\triangle ABC$ such that $PA = 3$, $PB = 4$, $PC = 5$. Find the side length of $\triangle ABC$. (Hint: rotate $\triangle APB$ by 60° about B .)

95. In a parallelogram $ABCD$, prove that $AC^2 + BD^2 = 2(AB^2 + AD^2)$. Use this and the Law of Cosines to show that, for any quadrilateral $ABCD$,

$$AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + DA^2 - 4 \cdot MN^2,$$

where M, N are the midpoints of the diagonals (*Euler's quadrilateral identity*).

96. A point P lies inside $\triangle ABC$. Prove that

$$PA^2 + PB^2 + PC^2 = GA^2 + GB^2 + GC^2 + 3 \cdot PG^2,$$

where G is the centroid (*Leibniz's identity*). Deduce that $PA^2 + PB^2 + PC^2$ is minimised at G .

97. Let H be the orthocentre of $\triangle ABC$. Prove that

$$HA^2 + BC^2 = HB^2 + CA^2 = HC^2 + AB^2 = 4R^2.$$

98. In $\triangle ABC$, the foot of the altitude from A to BC is D . Prove that $BD = c \cos B$, $DC = b \cos C$, and $AD = \frac{2[\triangle ABC]}{a}$. Use these to derive Heron's formula.

99. (*The Carnot identity*) Let O be the circumcentre of $\triangle ABC$, with d_a, d_b, d_c the signed distances from O to sides a, b, c . Prove $d_a + d_b + d_c = R + r$ for an acute triangle, and state the modification for an obtuse triangle.

100. In $\triangle ABC$, the altitudes from B and C have feet E, F . Prove $EF = a \cos A$ and that the orthic triangle $\triangle DEF$ has perimeter $a \cos A + b \cos B + c \cos C$. Conclude that $\triangle DEF$ is the triangle of least perimeter inscribed in an acute $\triangle ABC$ (*Fagnano's problem*).

Part 5 — Quadrilaterals

101. The diagonals of a convex quadrilateral $ABCD$ are perpendicular if and only if $AB^2 + CD^2 = BC^2 + DA^2$. Prove both directions.

102. A quadrilateral $ABCD$ has $AB + CD = BC + AD$. Prove that $ABCD$ admits an inscribed circle tangent to all four sides (*converse of Pitot*). Conversely, if $ABCD$ is tangential, prove $AB + CD = BC + AD$.

103. (*Varignon's Theorem and consequences*) In a quadrilateral $ABCD$, let P, Q, R, S be the midpoints of AB, BC, CD, DA . Prove $PQRS$ is a parallelogram with $[PQRS] = \frac{1}{2}[ABCD]$. Show further that $PQRS$ is a rectangle iff the diagonals of $ABCD$ are perpendicular, a rhombus iff the diagonals of $ABCD$ are equal, and a square iff both.

104. (*Brahmagupta's formula*) For a cyclic quadrilateral with sides a, b, c, d and semiperimeter $s = \frac{1}{2}(a + b + c + d)$,

$$[ABCD] = \sqrt{(s-a)(s-b)(s-c)(s-d)}.$$

Prove this and deduce Heron's formula as the limiting case $d \rightarrow 0$.

105. (*Bretschneider's formula*) For a general convex quadrilateral with sides a, b, c, d and opposite angles α, γ ,

$$[ABCD] = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd \cos^2\left(\frac{\alpha+\gamma}{2}\right)}.$$

106. A convex quadrilateral $ABCD$ has perpendicular diagonals meeting at E . Prove that $ABCD$ is cyclic if and only if $AE \cdot CE = BE \cdot DE$.

107. In a tangential polygon (one with an inscribed circle of radius r), prove that the area equals rs , where s is the semiperimeter. For a tangential quadrilateral, deduce $[ABCD] = \sqrt{abcd} \sin\left(\frac{\alpha+\gamma}{2}\right)$, where α, γ are opposite angles.

108. $ABCD$ is a convex quadrilateral. Prove that it is bicentric (cyclic and tangential) if and only if it is cyclic and $AB + CD = BC + AD$. For a bicentric quadrilateral, prove $[ABCD] = \sqrt{abcd}$ (*the bicentric area formula*).

109. (*Newton's Line*) The midpoints of the diagonals of any quadrilateral are collinear with the centroid of the four vertices. Prove this and prove further that for a tangential quadrilateral, the incentre also lies on this line (*Newton's theorem on tangential quadrilaterals*).

110. The diagonals of a cyclic quadrilateral $ABCD$ meet at P . Prove that

$$\frac{AP}{PC} = \frac{AB \cdot AD}{CB \cdot CD}, \quad \frac{BP}{PD} = \frac{BA \cdot BC}{DA \cdot DC}.$$

111. Let $ABCD$ be a cyclic quadrilateral. Prove that the four perpendiculars from the midpoint of one side to the opposite side, the midpoint of one diagonal to the opposite diagonal, etc., are concurrent. Identify this point as the *anticentre* of $ABCD$, and prove it is the reflection of the circumcentre over the centroid of the four vertices.

112. A convex quadrilateral $ABCD$ has $\angle ABC = \angle ADC = 90^\circ$. Prove $ABCD$ is cyclic with AC as diameter, and that the foot of the perpendicular from the intersection P of the diagonals to AC lies on the circle of diameter BD .

113. (*Japanese theorem for cyclic polygons*) For any cyclic quadrilateral $ABCD$, the incentres of $\triangle ABC, \triangle BCD, \triangle CDA, \triangle DAB$ are the vertices of a rectangle. Prove this.

114. (*The butterfly theorem*) Let M be the midpoint of a chord PQ of a circle. Two other chords AB and CD pass through M . The chords AD and BC meet PQ at X and Y . Prove that M is also the midpoint of XY .

115. (*IMO 1995 generalisation*) $ABCD$ is a cyclic quadrilateral. The diagonals meet at P . The circumcircles of $\triangle APB$ and $\triangle CPD$ meet again at Q . Prove that $\angle BQC = \angle AQD$, and that Q lies on line OP extended, where O is the centre of the circle through A, B, C, D .

Part 6 — Circle Geometry and Ptolemy's Theorem

116. (*Power of a Point*) From a point P in the plane of a circle ω , two secants are drawn meeting ω at A, B and C, D . Prove $PA \cdot PB = PC \cdot PD$ (with appropriate signs), and prove the analogous identity for a tangent from P touching ω at T : $PT^2 = PA \cdot PB$. Use this to characterise concyclicity of four points by the equality of two products.

117. (*Radical axis*) Given two circles, prove that the locus of points with equal power with respect to both is a line perpendicular to the line of centres. Given three circles in general position, prove the three radical axes concur at the *radical centre*.

118. Two circles intersect at A and B . A variable line through A meets the circles again at P and Q . Prove that the angle $\angle PBQ$ is constant.

119. Two circles touch internally at P . A chord AB of the larger touches the smaller at T . Prove PT bisects $\angle APB$, and that the locus of T as AB varies is the smaller circle itself.

120. The incircle of $\triangle ABC$ touches BC, CA, AB at D, E, F . Prove

$$AE = AF = s - a, \quad BD = BF = s - b, \quad CD = CE = s - c,$$

and deduce that $[\triangle ABC] = rs$. Generalise to the A -excircle and find the analogous contact-length formulas.

121. Prove that $abc = 4R \cdot [\triangle ABC]$, and deduce $\sin A \sin B \sin C = \frac{[\triangle ABC]}{2R^2}$. Conclude $\frac{a}{\sin A} = 2R$.

122. (*Euler's formula*) Prove $OI^2 = R^2 - 2Rr$ and deduce $R \geq 2r$, with equality iff $\triangle ABC$ is equilateral. Prove the corresponding excircle identity $OI_A^2 = R^2 + 2Rr_A$, where I_A is the A -excentre and r_A the corresponding exradius.

123. (*Ptolemy's Theorem*) $ABCD$ is a cyclic quadrilateral. Prove

$$AC \cdot BD = AB \cdot CD + AD \cdot BC.$$

124. (*Ptolemy's Inequality*) For any four points A, B, C, D in the plane,

$$AC \cdot BD \leq AB \cdot CD + AD \cdot BC,$$

with equality if and only if A, B, C, D are concyclic in this cyclic order.

125. P lies on the minor arc BC of the circumcircle of equilateral $\triangle ABC$. Use Ptolemy to prove $PA = PB + PC$, and prove the more general fact: if P lies on the arc BC of the circumcircle of $\triangle ABC$ not containing A , then $\frac{PA}{a} = \frac{PB}{b} + \frac{PC}{c}$ holds for an isosceles triangle iff a corresponds to the apex side.

126. (*Generalised Ptolemy / Casey's Theorem*) Four circles $\omega_1, \omega_2, \omega_3, \omega_4$ are tangent internally to a circle Ω in this cyclic order. Let t_{ij} denote the length of the common external tangent of ω_i, ω_j . Prove

$$t_{12}t_{34} + t_{14}t_{23} = t_{13}t_{24}.$$

127. Use Ptolemy's theorem on the cyclic quadrilateral inscribed in a regular pentagon $ABCDE$ to derive $\frac{AC}{AB} = \frac{1+\sqrt{5}}{2}$ (the golden ratio), and use the same technique on a regular heptagon to derive the identity

$$\frac{1}{a_1} = \frac{1}{a_2} + \frac{1}{a_3},$$

where $a_1 < a_2 < a_3$ are the three distinct chord lengths in a regular heptagon.

128. Let Γ be the circumcircle of $\triangle ABC$. The tangent to Γ at A meets line BC at T . Prove $TA^2 = TB \cdot TC$, $\frac{TB}{TC} = \frac{c^2}{b^2}$, and that the tangent at A , the tangent at the antipode of A , and side BC are concurrent or parallel.

129. Let the incircle of $\triangle ABC$ touch BC at X . Let X' be the point diametrically opposite X on the incircle. Prove that A, X' , and the contact point of the A -excircle with BC are collinear.

130. The contact triangle (*Gergonne triangle*) and the medial triangle of $\triangle ABC$ are perspective from a point. Prove this point lies on the line joining the incentre to the centroid, and that the corresponding Cevians from A, B, C to the contact points concur at the *Gergonne point*. State the analogous result for the Nagel point.

131. (*Mid-arc / Incentre-Excentre Lemma*) Let M be the midpoint of arc BC of the circumcircle of $\triangle ABC$ not containing A . Prove $MB = MC = MI = MI_A$, where I, I_A are the incentre and the A -excentre. Conclude that B, C, I, I_A lie on a circle centred at M .

132. (*Nine-Point Circle*) Prove that the midpoints of the three sides, the feet of the three altitudes, and the midpoints of the segments from each vertex to the orthocentre lie on a common circle of radius $R/2$, with centre at the midpoint of OH on the Euler line.

133. (*Feuerbach's Theorem*) Prove that the nine-point circle is tangent internally to the incircle and externally to each of the three excircles.

134. The reflections of the orthocentre H of $\triangle ABC$ in the three sides lie on the circumcircle of $\triangle ABC$. The reflection of H in the midpoint of a side is the antipode of the opposite vertex. Prove both, and deduce that the locus of H as A varies on a fixed circle (with B, C fixed) is a circle congruent to the circumcircle.

135. (*Simson Line*) Let P lie on the circumcircle of $\triangle ABC$. The feet of perpendiculars from P to the three sides are collinear. Prove this, prove the converse, and prove that as P moves around the circumcircle, the Simson line of P rotates at half the angular rate of P .

136. (*Steiner Line*) The reflections of P (on the circumcircle) in the three sides of $\triangle ABC$ are collinear, and this line passes through the orthocentre H . Prove this and prove that the Steiner line is parallel to the Simson line, at twice the distance from P .

137. (*Pedal Triangle area*) For a point P with pedal triangle $\triangle DEF$ in $\triangle ABC$, prove

$$[\triangle DEF] = \frac{|R^2 - OP^2|}{4R^2} [\triangle ABC],$$

where O is the circumcentre. Deduce that the pedal triangle degenerates iff P lies on the circumcircle (Simson line).

138. (*Miquel's Theorem*) On the sides BC, CA, AB of $\triangle ABC$, choose D, E, F . Prove the circumcircles of $\triangle AEF, \triangle BFD, \triangle CDE$ have a common point (*Miquel point*). Prove furthermore that this point lies on the circumcircle of $\triangle ABC$ iff D, E, F are collinear.

139. (*Miquel point of a complete quadrilateral*) Four lines in general position determine four triangles. Prove the four circumcircles share a common point, and that this point lies on the circle through the four diagonal-vertex orthocentres.

140. (*Reim's Theorem*) Two circles meet at A and B . Lines through A and B meet the circles again at P, P' and Q, Q' respectively. Prove $PQ \parallel P'Q'$. State and prove the converse.

Part 7 — Ceva's and Menelaus' Theorems

141. (*Ceva*) Cevians AD, BE, CF of $\triangle ABC$ are concurrent if and only if $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$ (signed). Prove both directions.

142. (*Trigonometric Ceva*) The cevians from A, B, C are concurrent if and only if

$$\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{\sin \angle CBE}{\sin \angle EBA} \cdot \frac{\sin \angle ACF}{\sin \angle FCB} = 1.$$

143. (*Menelaus*) A transversal meets lines BC, CA, AB at D, E, F . Prove $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1$ (signed). Prove the converse.

144. (*The cevian-nest theorem / Ceva nested*) If cevians AD, BE, CF of $\triangle ABC$ are concurrent at P , and cevians of $\triangle DEF$ from each vertex through the corresponding side of $\triangle ABC$ are concurrent, then the second concurrence point lies on a fixed line through P . Prove this.

145. P is a point inside $\triangle ABC$, and AP, BP, CP meet the sides at D, E, F . Prove

$$\frac{PD}{AD} + \frac{PE}{BE} + \frac{PF}{CF} = 1, \quad \frac{AP}{AD} + \frac{BP}{BE} + \frac{CP}{CF} = 2.$$

146. A line through the centroid G of $\triangle ABC$ meets sides AB, AC at P, Q . Prove $\frac{AB}{AP} + \frac{AC}{AQ} = 3$, and characterise the lines through G for which P, Q both lie on *segments* (rather than extensions).

147. In $\triangle ABC$, three concurrent cevians AD, BE, CF meet at P . Prove

$$\frac{AF}{FB} + \frac{AE}{EC} = \frac{AP}{PD},$$

and find the analogous identities at the other two vertices.

148. (*Routh's Theorem*) On the sides BC, CA, AB of $\triangle ABC$, points D, E, F are taken with $\frac{BD}{DC} = x, \frac{CE}{EA} = y, \frac{AF}{FB} = z$. Prove that the cevians AD, BE, CF form an inner triangle whose ratio of area to $[\triangle ABC]$ is

$$\frac{(xyz - 1)^2}{(xy + y + 1)(yz + z + 1)(zx + x + 1)}.$$

Verify the famous $\frac{1}{7}$ ratio when $x = y = z = 2$.

149. (*Desargues' Theorem*) Two triangles $\triangle ABC$ and $\triangle A'B'C'$ are perspective from a point if lines AA', BB', CC' concur. Prove they are then perspective from a line, i.e. $BC \cap B'C', CA \cap C'A', AB \cap A'B'$ are collinear. Prove the converse.

150. (*Pappus' Theorem*) Six points A_1, A_2, A_3 on one line and B_1, B_2, B_3 on another. Let $P_{ij} = A_i B_j \cap A_j B_i$. Prove P_{12}, P_{13}, P_{23} are collinear (the *Pappus line*).

151. (*Pascal's Theorem*) Six points A, B, C, D, E, F on a conic. The three points $AB \cap DE, BC \cap EF, CD \cap FA$ are collinear. Use this to prove that the three pairs of opposite sides of a hexagon inscribed in a circle meet on a single line, and derive Pappus as a degenerate case.

152. (*Brianchon's Theorem*) The three main diagonals of a hexagon circumscribed about a conic are concurrent. Prove this and use it to prove that, in a tangential hexagon $AA'BB'CC'$ inscribed about a circle, the diagonals AB', BC', CA' are concurrent.

153. (*Newton–Gauss Line*) In a complete quadrilateral, the midpoints of the three diagonals are collinear. Prove this using Menelaus, and identify the line as the Newton–Gauss line of the quadrilateral.

154. In $\triangle ABC$, the symmedian from A meets BC at K . Prove $\frac{BK}{KC} = \frac{c^2}{b^2}$, and that the three symmedians concur at the *symmedian point* (Lemoine point) K , with $\overrightarrow{AK} \parallel \overrightarrow{a^2 \mathbf{u}_a + b^2 \mathbf{u}_b + c^2 \mathbf{u}_c}$ for appropriate vectors.

155. (*Cevian projection lemma*) From a point P inside $\triangle ABC$, cevians AD, BE, CF are drawn. Let D', E', F' be the reflections of D, E, F in the midpoints of the corresponding sides. Prove that AD', BE', CF' are also concurrent (at the *isotomic conjugate* of P).

Part 8 — Triangle Centres and Advanced Topics

156. (*Euler Line*) Prove that the circumcentre O , centroid G , and orthocentre H are collinear, with $OG : GH = 1 : 2$, and that $\overrightarrow{OH} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$.

157. (*Orthocentric system*) The four points A, B, C, H form a system in which each is the orthocentre of the triangle of the other three. Prove this and prove that the four triangles share a common nine-point circle.

158. Prove

$$OH^2 = R^2 - 8R^2 \cos A \cos B \cos C, \quad IH^2 = 2r^2 - 4R^2 \cos A \cos B \cos C,$$

and use these to prove that $\cos A \cos B \cos C \leq \frac{1}{8}$, with equality iff $\triangle ABC$ is equilateral.

159. (*Identities of triangle invariants*) Prove

$$\cos A + \cos B + \cos C = 1 + \frac{r}{R}, \quad \sin A + \sin B + \sin C = \frac{s}{R},$$

$$\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1.$$

160. (*Heron via the incircle*) Prove $[\triangle ABC] = (s - a) \tan \frac{A}{2} \cdot s = \sqrt{s(s - a)(s - b)(s - c)} = rs = \frac{abc}{4R}$, exhibiting the equivalence of all four expressions.

161. (*Napoleon's Theorem, with refinement*) Equilateral triangles are constructed externally on the sides of any $\triangle ABC$. Their centres form an equilateral triangle (the *outer Napoleon triangle*). Prove this, and prove that the difference of areas of the outer and inner Napoleon triangles equals $[\triangle ABC]$.

162. (*Fermat–Torricelli Point*) For a triangle whose every angle is less than 120° , prove there is a unique point F inside it minimising $FA + FB + FC$, and that $\angle AFB = \angle BFC = \angle CFA = 120^\circ$. For $\angle A \geq 120^\circ$, prove the minimiser is the vertex A . Identify F as the intersection of three Napoleon-type lines.

163. (*Morley's Theorem*) The three pairwise intersections of adjacent angle trisectors of any triangle form an equilateral triangle (the *Morley triangle*). Prove this. State the analogous result for the external trisectors.

164. (*Erdős–Mordell Inequality*) Let P be a point inside $\triangle ABC$ with distances d_a, d_b, d_c to the sides and R_a, R_b, R_c to the vertices. Prove

$$R_a + R_b + R_c \geq 2(d_a + d_b + d_c),$$

with equality iff $\triangle ABC$ is equilateral and P is its centre. State and prove the multiplicative refinement: $R_a R_b R_c \geq 8 d_a d_b d_c$.

165. (*Isogonal Conjugates*) Two points P and Q are isogonal conjugates in $\triangle ABC$ if at each vertex, the cevians to P and Q are reflections in the angle bisector. Prove the orthocentre and circumcentre are isogonal conjugates, and the incentre is its own isogonal conjugate. Prove that the pedal triangles of P and Q are inscribed in a common circle (the *common pedal circle*) centred at the midpoint of PQ .

166. (*Isotomic Conjugates*) The reflections of D, E, F (where the cevians from A, B, C meet the opposite sides) in the midpoints of those sides give cevians whose lines concur at the *isotomic conjugate* of P . Prove this and identify the isotomic conjugates of the centroid, the symmedian point, and the Gergonne point.

167. (*Apollonius Circle*) The locus of points P with $\frac{PB}{PC} = k$ (a fixed positive real $\neq 1$) is a circle through the two points dividing BC internally and externally in the ratio $k : 1$. Prove this. For $k = c/b$, prove the Apollonius circle passes through A and is preserved under reflection in the A -symmedian.

168. (*Three Apollonius circles*) The three Apollonius circles of $\triangle ABC$ (corresponding to ratios $c/b, a/c, b/a$ at the three vertices) have a common axis perpendicular to the Euler line, and they meet pairwise in two points (the *isodynamic points*). Prove these statements and prove the isodynamic points are isogonal conjugates of the Fermat points.

169. (*Spiral Similarity Lemma*) Let $\triangle PAB$ and $\triangle PCD$ be directly similar, sharing the vertex P . Prove that $\triangle PAC$ and $\triangle PBD$ are also directly similar, with the same centre P . Apply this to prove: if $ABCD$ is a quadrilateral and $X = AC \cap BD$, $Y = AB \cap CD$, then the spiral similarity sending AB to DC has centre at the second intersection of circles (YAD) and (YBC) .

170. (*Inversion — properties*) Let ω be a circle with centre O and radius r . Define the inverse of $P \neq O$ as the point P^* on ray OP with $OP \cdot OP^* = r^2$. Prove that: (a) lines through O map to themselves; (b) lines not through O map to circles through O ; (c) circles through O map to lines not through O ; (d) circles not through O map to circles not through O ; (e) the angle between two curves is preserved under inversion.

171. Use inversion centred at A to give a one-line proof of Ptolemy's inequality, and use inversion centred at the incentre I of $\triangle ABC$ to prove that the circumcircles of $\triangle BIC$, $\triangle CIA$, $\triangle AIB$ are mutually congruent (each of radius $\frac{a}{2 \cos(A/2)}$ etc.).

172. (*Weitzenböck and Hadwiger–Finsler*) Prove

$$a^2 + b^2 + c^2 \geq 4\sqrt{3} [\triangle ABC] + (a-b)^2 + (b-c)^2 + (c-a)^2,$$

with equality iff $\triangle ABC$ is equilateral, and deduce the (weaker) Weitzenböck inequality $a^2 + b^2 + c^2 \geq 4\sqrt{3} [\triangle ABC]$.

173. (*Euler's Quadrilateral / Carnot-style identity*) Let O be the circumcentre and G the centroid of $\triangle ABC$. Prove

$$OG^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2).$$

Deduce $a^2 + b^2 + c^2 \leq 9R^2$, with equality iff $\triangle ABC$ is equilateral.

174. Prove that for any point P in the plane of $\triangle ABC$,

$$PA^2 \sin 2A + PB^2 \sin 2B + PC^2 \sin 2C = a^2 \sin 2A \cdot \frac{[\triangle PBC]}{[\triangle ABC]} + \dots$$

(barycentric distance formula). Apply to $P = O$ (circumcentre) and $P = I$ (incentre) to recover $OA = OB = OC = R$ and Euler's $OI^2 = R^2 - 2Rr$.

175. (*Kiepert hyperbola*) Isosceles triangles with apex angle θ are erected on the sides of $\triangle ABC$ (all externally or all internally). The lines joining each apex to the opposite vertex of $\triangle ABC$ are concurrent at a point $K(\theta)$. As θ varies, $K(\theta)$ traces a conic. Prove this conic is a rectangular hyperbola passing through the centroid, orthocentre, and the two Fermat points.

176. (*Lemoine / symmedian properties*) Prove that the symmedian point K of $\triangle ABC$ minimises $PA^2 + PB^2 + PC^2$ subject to a constraint relating the distances of P to the three sides. Prove also that the pedal triangle of K has centroid coinciding with K , and that K is the unique point with this property.

177. (*IMO 1996 / Tangent-length identity*) In $\triangle ABC$, let the angle bisector from A meet BC at D and the circumcircle at E . Prove

$$AD \cdot AE = AB \cdot AC, \quad AD^2 = AB \cdot AC - BD \cdot DC, \quad DE = \frac{a^2}{4R} \cdot \frac{1}{\cos(A/2)}.$$

178. (*The orthopole*) Given a line ℓ and a triangle $\triangle ABC$, drop perpendiculars from A, B, C to ℓ with feet A', B', C' . From A', B', C' drop perpendiculars to the sides BC, CA, AB respectively. Prove these three perpendiculars are concurrent at a point (the *orthopole* of ℓ). Prove furthermore that as ℓ rotates about a fixed point, its orthopole traces a conic.

179. (*Concluding inequality / IMO Shortlist*) Prove that for any $\triangle ABC$ with the usual notation,

$$27R^2 \geq (a+b+c)^2, \quad \frac{1}{a+b-c} + \frac{1}{b+c-a} + \frac{1}{c+a-b} \geq \frac{1}{a} + \frac{1}{b} + \frac{1}{c},$$

and characterise equality in each.